

P1.6 Energy dissipation

Learning objectives

After this topic, you should know:

- what is meant by useful energy
- what is meant by wasted energy
- what eventually happens to wasted energy
- whether energy is still as useful after it is used.

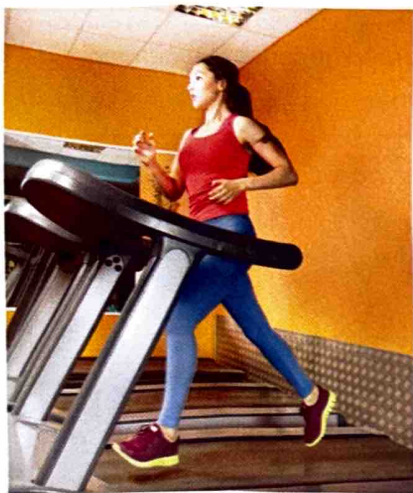


Figure 1 Using energy on a running machine

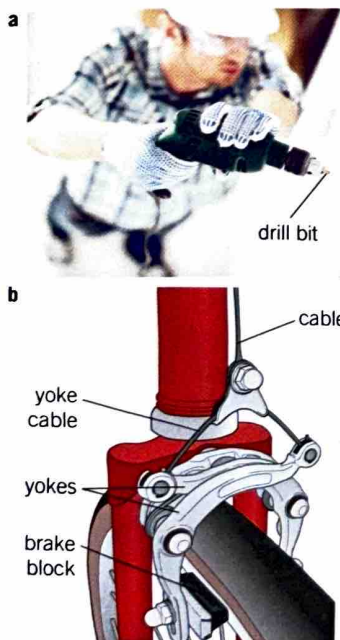


Figure 2 Friction in action: **a** Using a drill, **b** Braking on a bicycle

Energy for a purpose

Where would people be without machines? Washing machines are used to clean your clothes. Machines in factories are used to make the things you buy. You might use exercise machines in a gym to keep yourself fit, and machines are used to get you from place to place.

A machine transfers energy for a purpose. Friction between the moving parts of a machine causes the parts to warm up. So, not all of the energy supplied to a machine is usefully transferred. Some of the energy is wasted.

- **Useful energy** is energy transferred to where it is wanted in the way that is wanted.
- **Wasted energy** is the energy that is not usefully transferred.

Whenever energy is transferred for a purpose in any system, some of the energy is transferred usefully. The rest is **dissipated** (spreads out) and may be stored in less useful ways. This energy is described as wasted energy because it is not transferred as useful energy. For example, when a jet plane takes off, its engines transfer energy from the chemical energy store in the fuel. Some of this energy increases the kinetic energy and the gravitational potential energy stores of the plane, which is the useful energy transfer pathway. The rest is wasted energy because some of it heats the plane and the surroundings and some is transferred to the surroundings by sound waves created by the engine vibrations.

Investigating friction

Friction in machines always causes energy to be wasted. Figure 2 shows two examples of friction in action.

In Figure 2a, friction acts between the drill bit and the wood. The bit becomes hot as it bores into the wood. Energy is transferred by an electric current to the thermal energy stores, heating up the drill bit and the wood.

When a bike rider brakes, friction acts between the brake blocks and the wheels (Figure 2b). Energy is wasted as it is transferred from the kinetic energy stores of the bike and the cyclist, to the thermal energy stores of the brake blocks and the wheels, which are heated by friction.

Friction at work

Next time you use a running machine, think about what happens to the energy transferred by your muscles. As you are exercising, energy is transferred to the thermal energy store of your muscles (so you get hot) and to the thermal energy store of the machine by the force you exert to overcome the friction in the machine.

Spreading out

- Wasted energy is dissipated (spreads out) to the surroundings, for example, the gears of a car get hot because of friction when the car is running. Here, energy is transferred from the kinetic energy store of the gear box to the thermal energy stores of the gear box and the surrounding air. The thermal energy stores of the gear box and the surrounding air therefore increase, as do their respective temperatures.
- Useful energy is eventually transferred to the surroundings too. For example, the useful energy supplied to turn the wheels of a car is eventually transferred from the kinetic energy stores of the wheels to the thermal energy stores of the tyres by heating – increasing the thermal energy stores of the tyres. This energy is then transferred to the thermal energy store of the road and surrounding air.
- Energy becomes less useful the more it spreads out. For example, the hot water from a central heating boiler in a building is pumped through pipes and radiators. The thermal energy store of the hot water decreases as it transfers energy by heating to the thermal energy stores of the radiators – heating the rooms in the building. But the energy supplied to heat these rooms will eventually be transferred to the surrounding air.

1 Copy and complete the table below.

Energy transfer by	Useful energy output	Wasted energy output
a An electric fan heater	warms the air and surrounding objects	
b A television		
c An electric kettle		
d Headphones		

[4 marks]

2 Describe what would happen, in terms of energy transfer and changes to energy stores, to:

- a a gear box that was insulated so it could not transfer energy by heating to the surroundings [3 marks]
- b the running shoes of a jogger if the shoes are well insulated [2 marks]
- c a blunt electric drill bit if you use it to drill into hard wood [2 marks]
- d the metal wheel discs of the brakes of a car when the brakes are applied. [2 marks]

3 a Describe the energy transfers and changes to energy stores of a pendulum as it swings from one side to the middle, and then to the opposite side.  [4 marks]

b Explain why a swinging pendulum eventually stops. [3 marks]

4 A freewheeling cyclist on a level road gradually stops moving. Describe the energy transfers and changes to the energy stores of the cyclist and the bicycle.  [4 marks]

Key points

- Useful energy is energy in the place we want it and in the form we need it.
- Wasted energy is the energy that is not useful energy and is transferred by an undesired pathway.
- Wasted energy is eventually transferred to the surroundings, which become warmer.
- As energy dissipates (spreads out), it gets less and less useful.